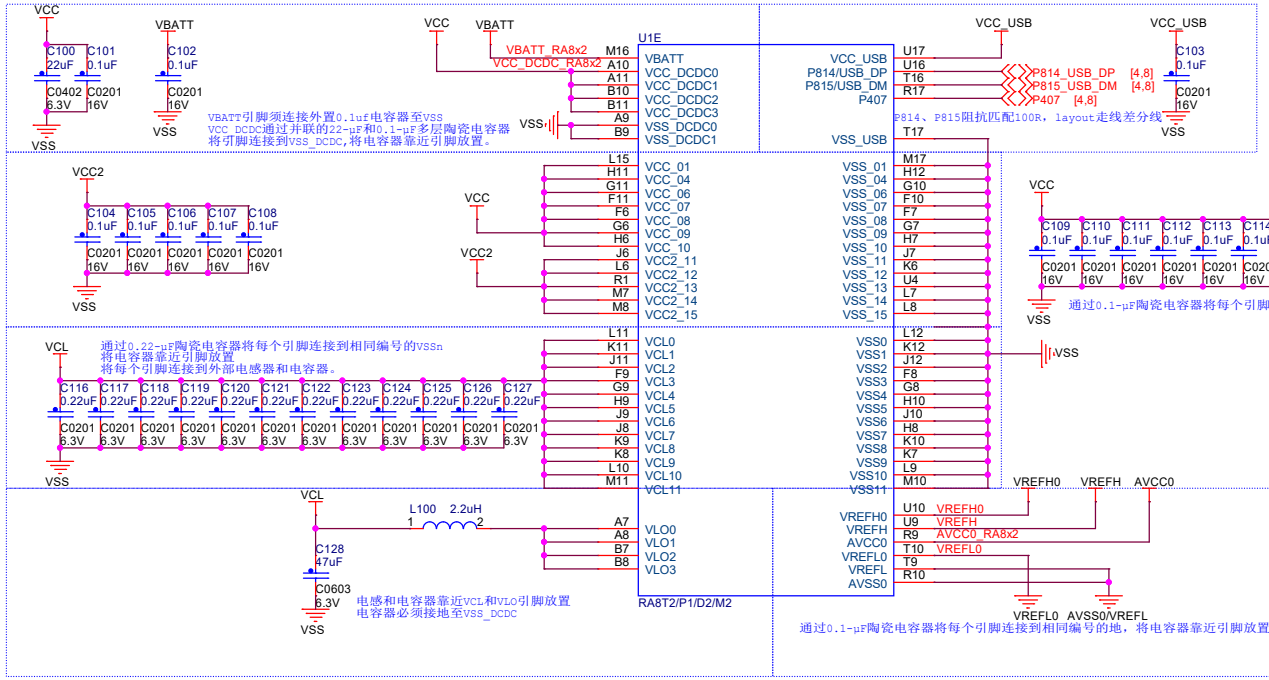
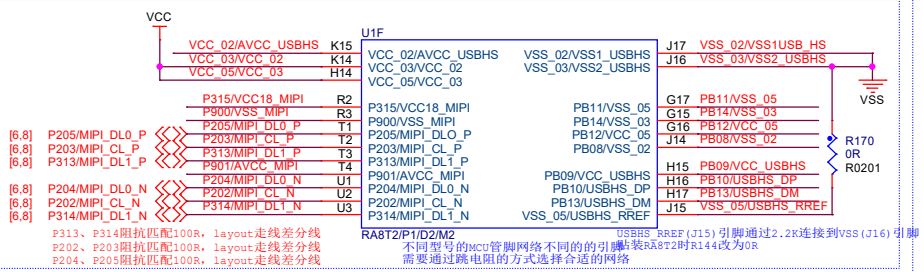
Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



核心板缺省功能阻容设置

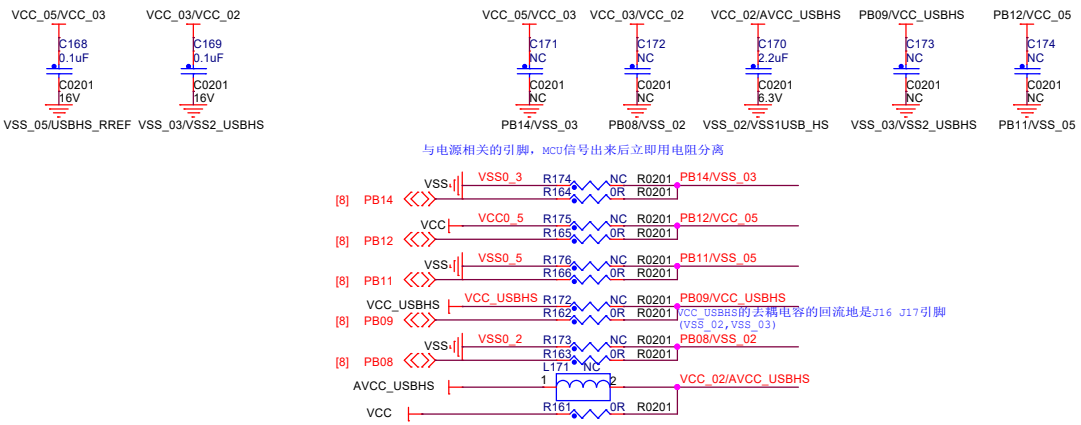
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1UF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHZ
R172-R180	NC	0R
R191-R193	NC	0R



由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2

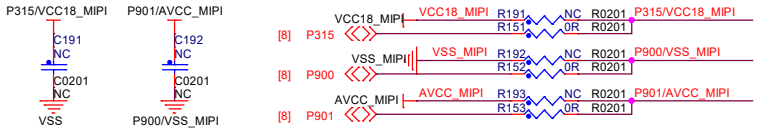
电阻选装

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



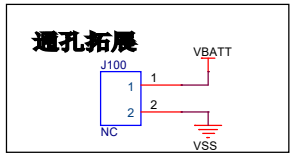
MIPI部分

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放





通孔拓展



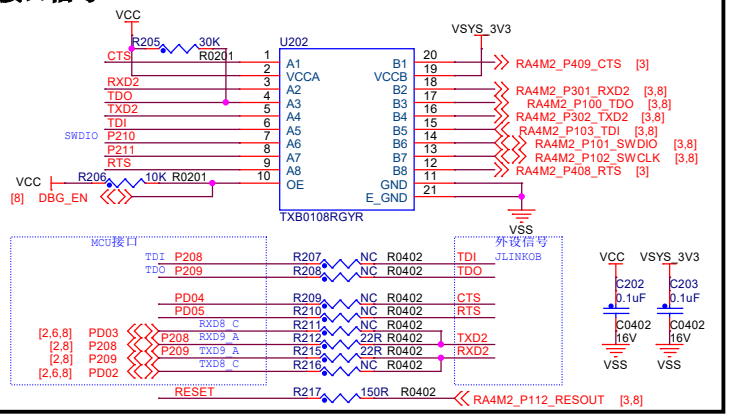
东莞野火电子技术有限公司 https://yehuosm.tmall.com			
Title: 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图			
Size: A3	Document Number: MCU 电源		Rev: V1.0
Date: Tuesday, June 09, 2026		Sheet 1 of 10	

Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

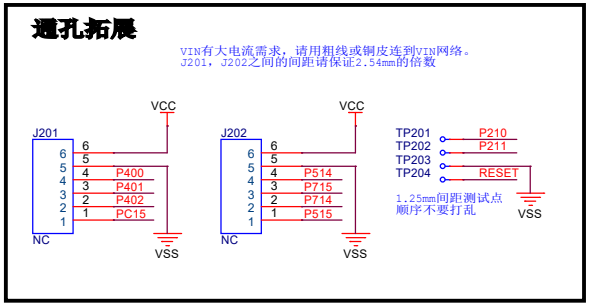
用户按键

LED

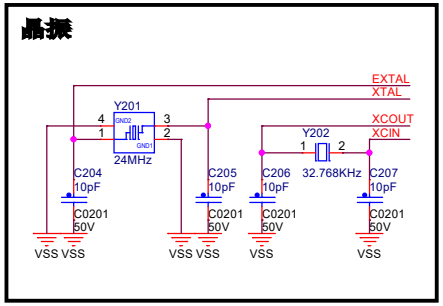
接口信号



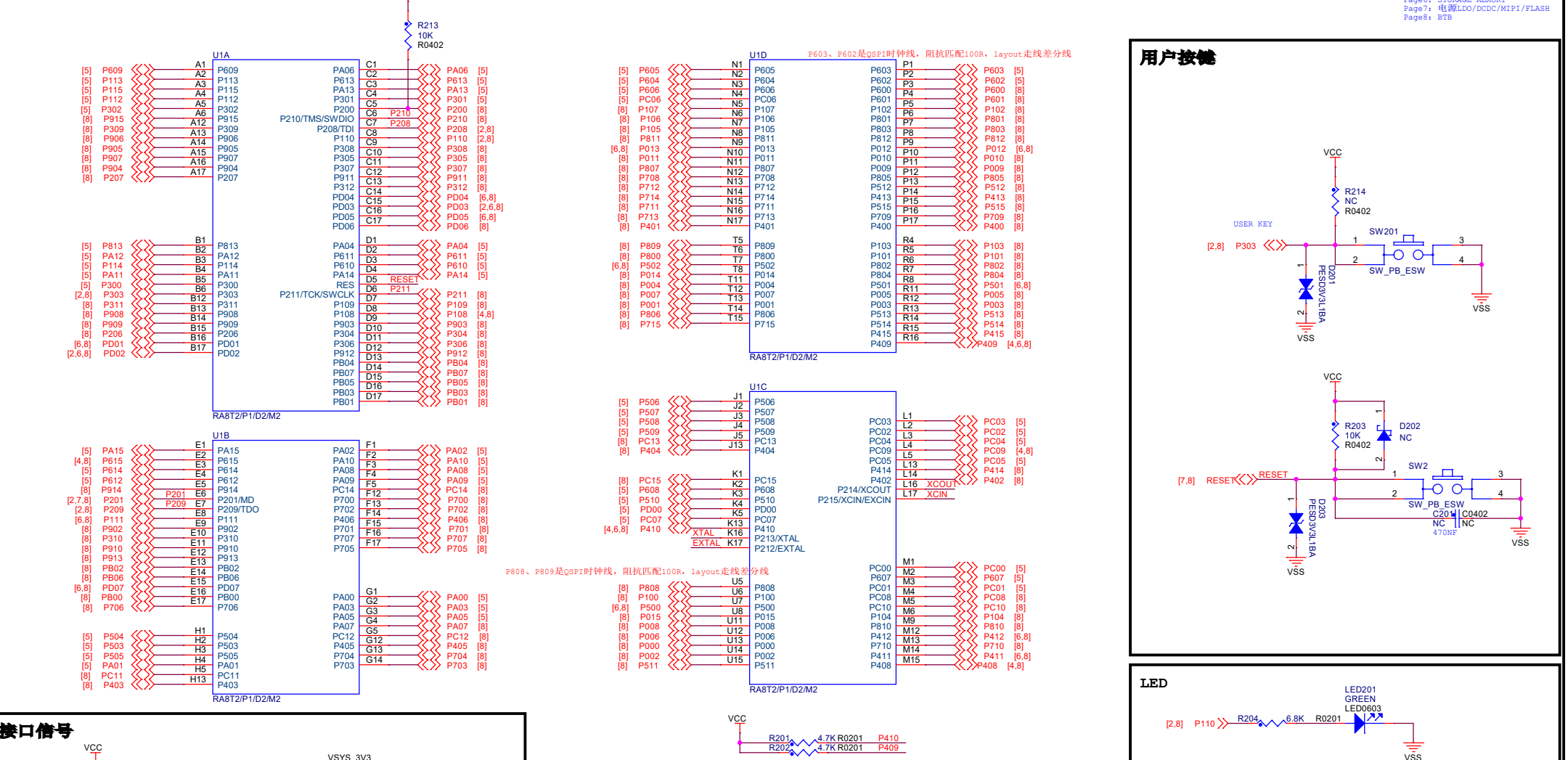
通孔拓展



晶振



如果使用USB HS，主晶振只能选12/20/24/48MHz



东莞野火电子技术有限公司
https://yehuosm.tmall.com

Title	野火瑞萨RA8T2-BTB核心板(PMIC)_原理图	
Size A3	Document Number MCU IO	Rev V1.0
Date: Tuesday, June 09, 2026	Sheet 2 of 10	

Jlink OB

注意：R7FA4M2只能12MHz晶振

Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

MCU调试/下载接口

[2,8] RA4M2_P302_TXD2
[2,8] RA4M2_P301_RXD2
[2,8] RA4M2_P100_TDO
[2,8] RA4M2_P101_SWDIO
[2,8] RA4M2_P102_SWCLK
[2,8] RA4M2_P103_TDI
JLOB_RESOUT

U5
R7FA4M2AD3CNE

MD	MODE
1	Single-chip mode
0	SCI/USB boot mode

layout：USB差分线控阻抗90R：
DBG_USB_DP/DBG_USB_DM

测试点 1.25间距/预留

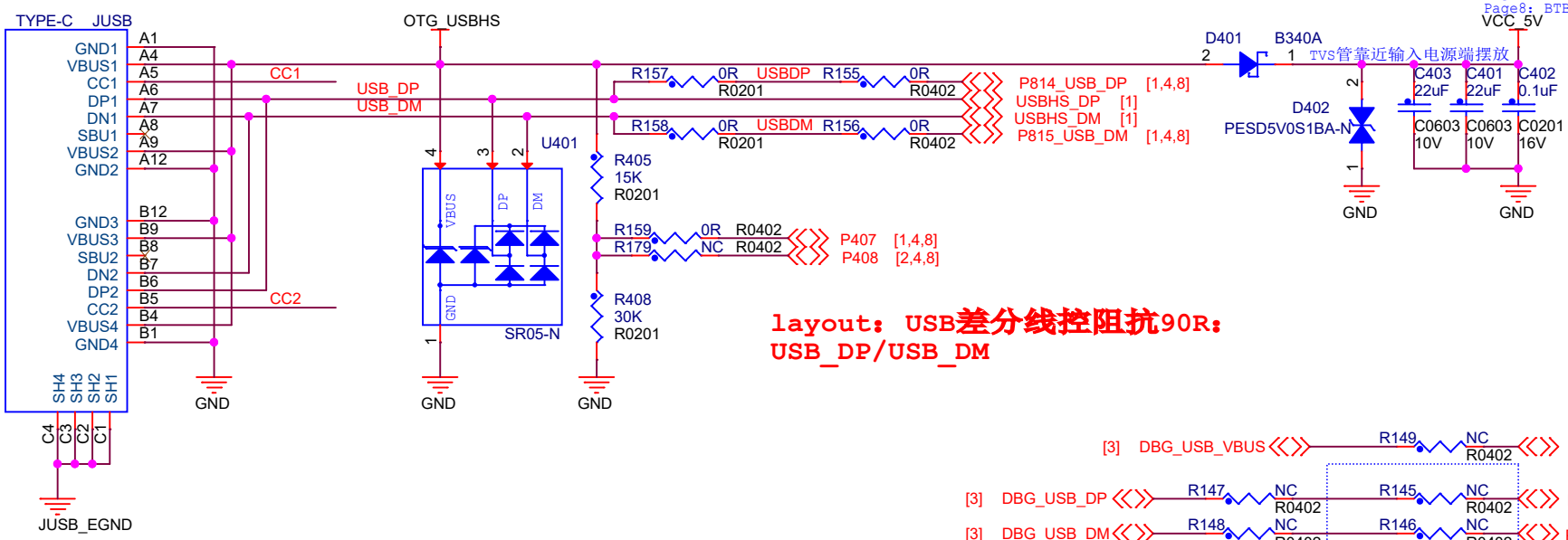
debug隔离工作时，用作短路二极管运行

东莞野火电子技术有限公司	https://yehuosm.tmall.com
Title	野火瑞萨RA8T2-BTB核心板(PMIC)_原理图
Size A4	Document Number Jlink OB
Rev V1.0	
Date: Tuesday, June 09, 2026	Sheet 3 of 10

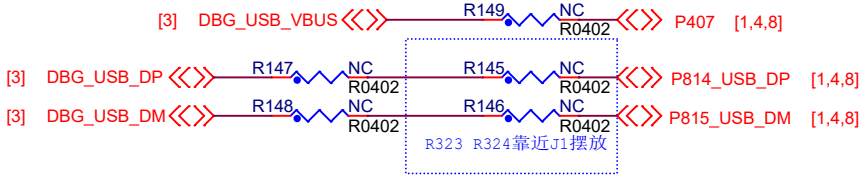
USB HS

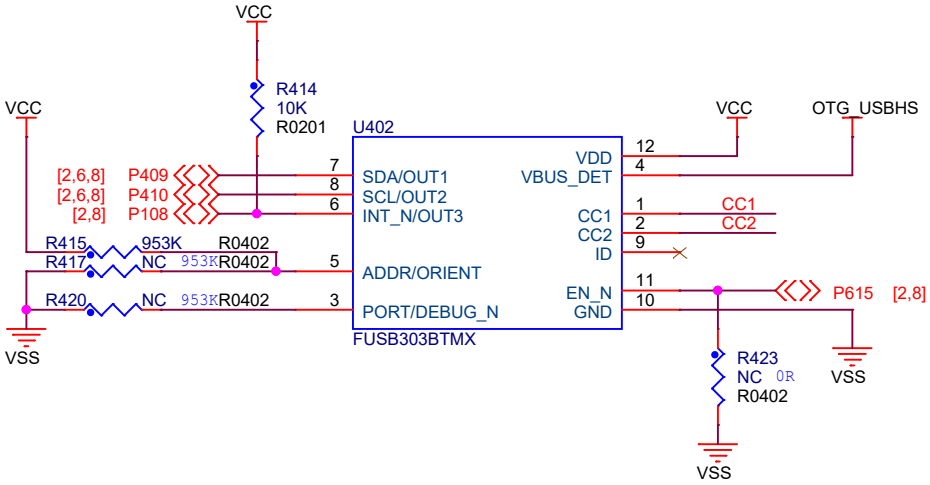
如果使用USB HS，主晶振只能选12/20/24/48MHz

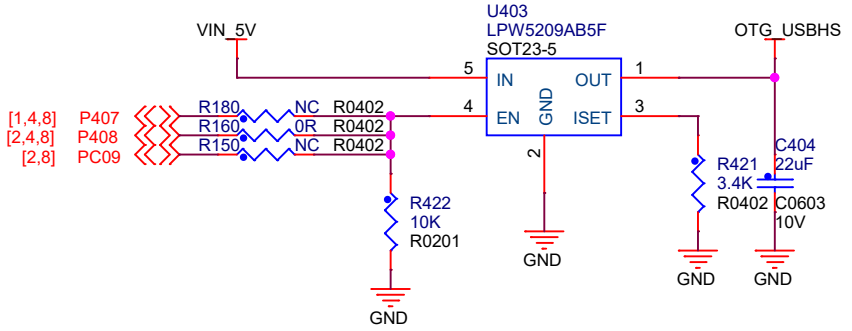
Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



layout：USB差分线控阻抗90R：
USB_DP/USB_DM

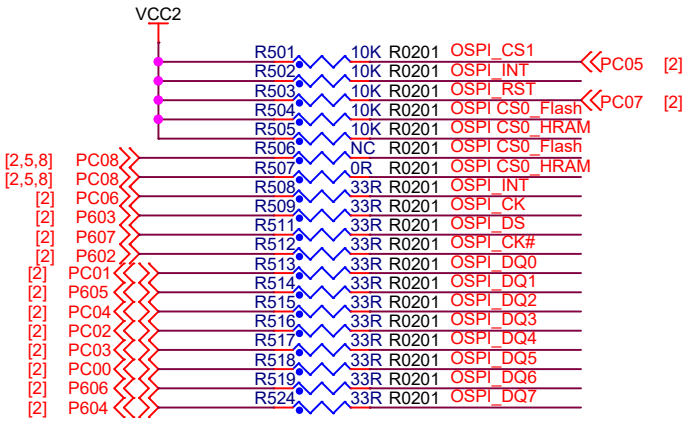


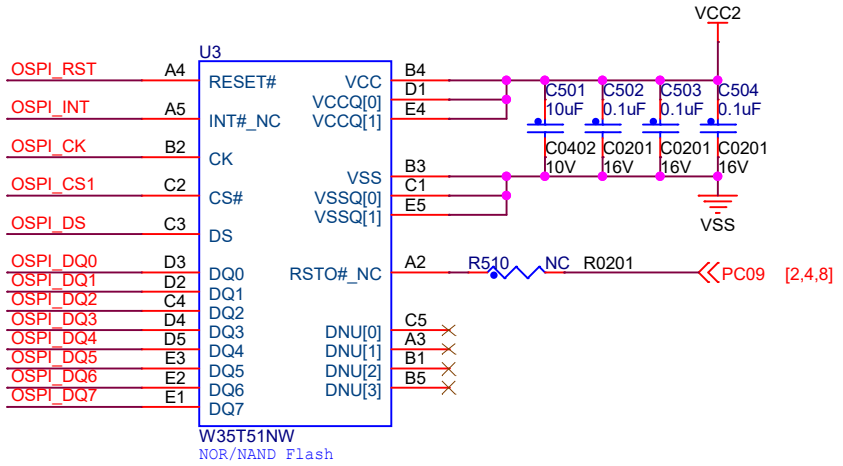




东莞野火电子技术有限公司 https://yehuosm.tmall.com		
Title: 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图		
Size: A4	Document Number: USB HS	Rev: V1.0
Date: Monday, June 22, 2026	Sheet 4 of 10	

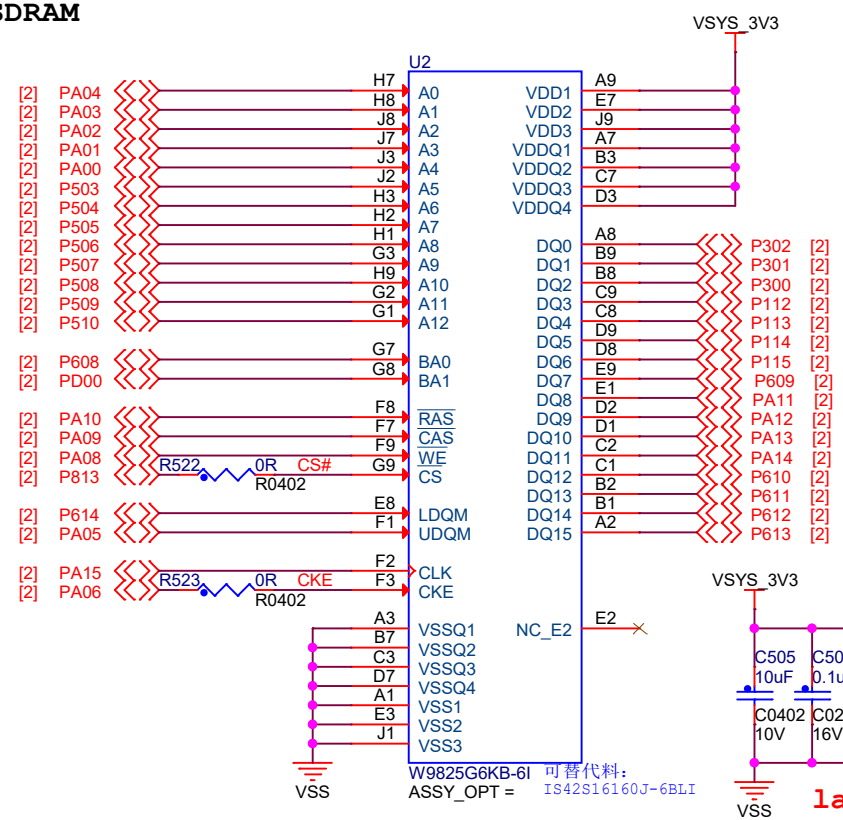
FLASH

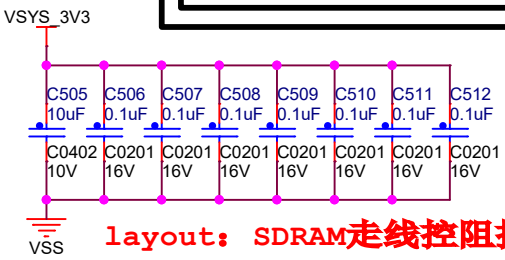


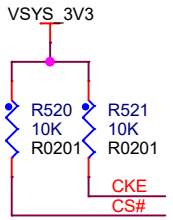


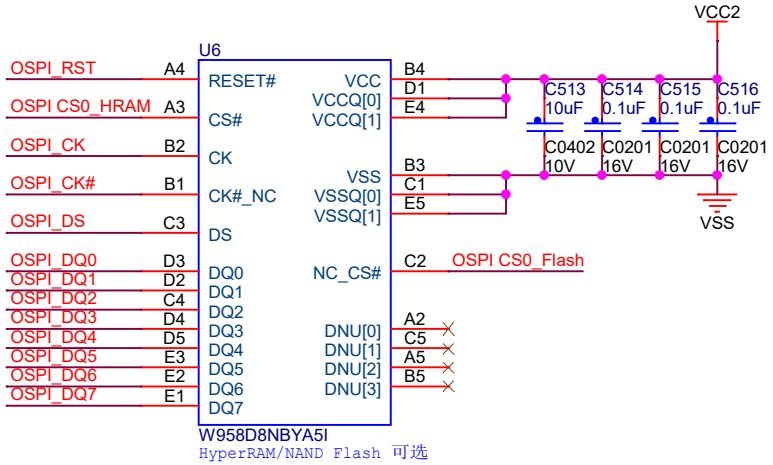
Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

SDRAM







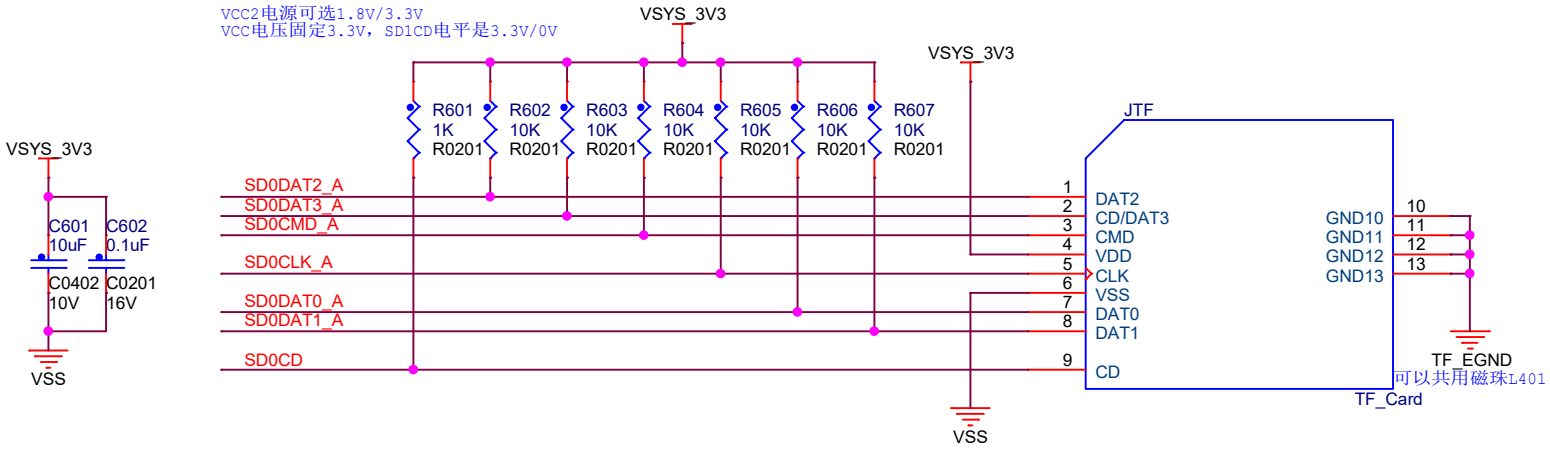


layout: SDRAM走线控阻抗50R

东莞野火电子技术有限公司
https://yehuosm.tmall.com

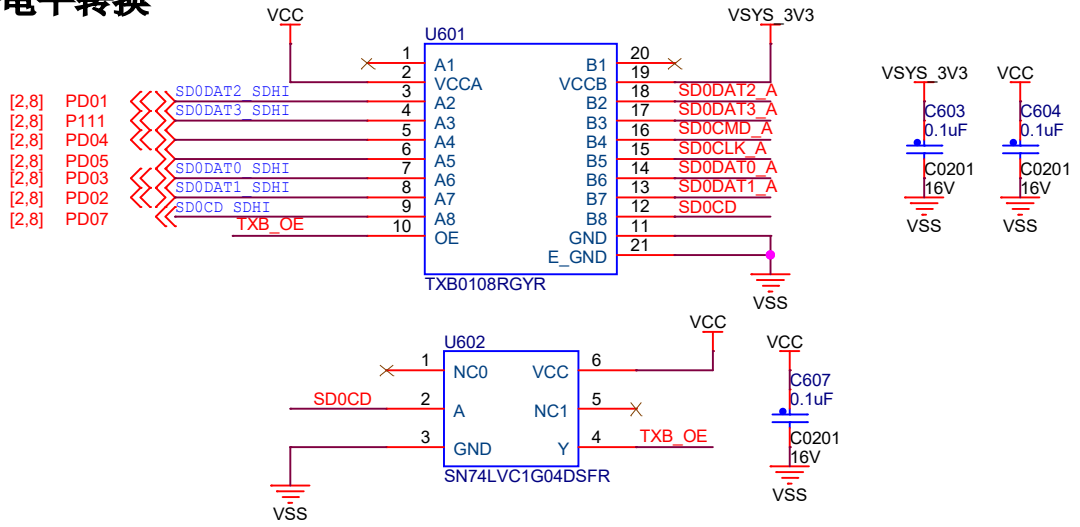
Title: 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图		
Size: A4	Document Number: SDRAM/FLASH	Rev: V1.0
Date: Tuesday, June 09, 2026	Sheet 5 of 10	

SD卡

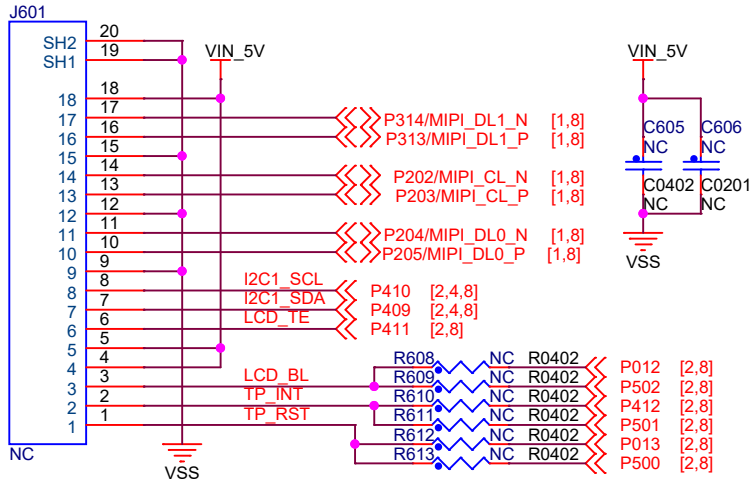


Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

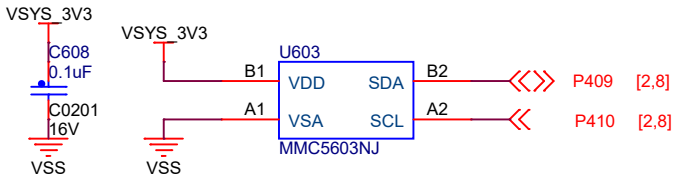
SD卡电平转换



MIPI屏

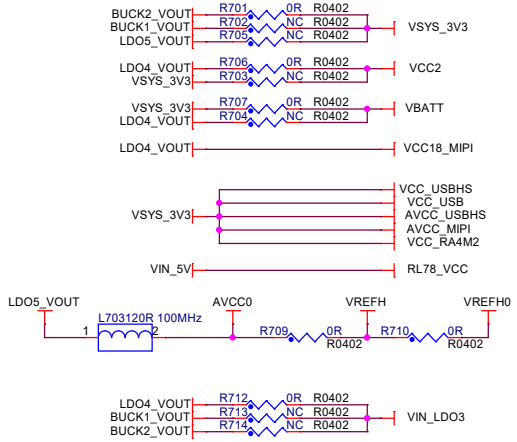


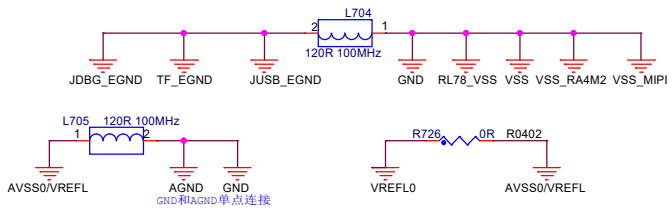
地磁传感器



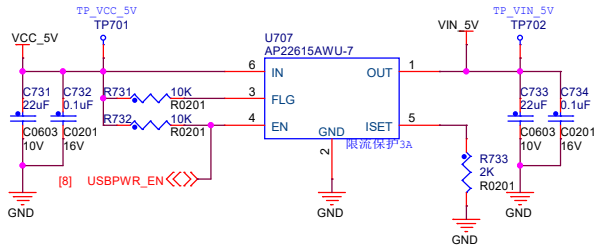
东莞野火电子技术有限公司 https://yehuosm.tmall.com		
Title: 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图		
Size: A4	Document Number: SD CARD/MIPI	Rev: V1.0
Date: Tuesday, June 09, 2026	Sheet 6 of 10	

电源分割

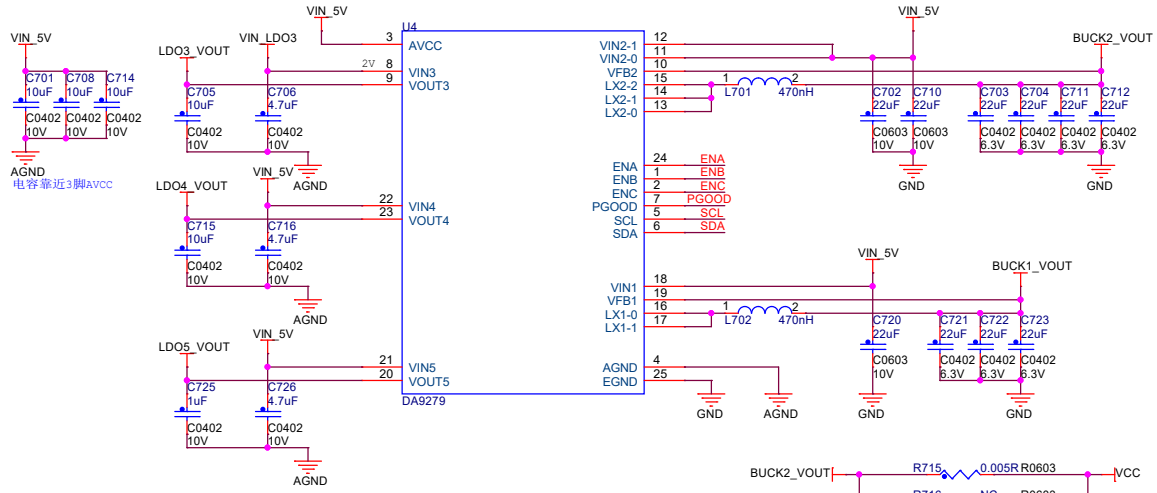




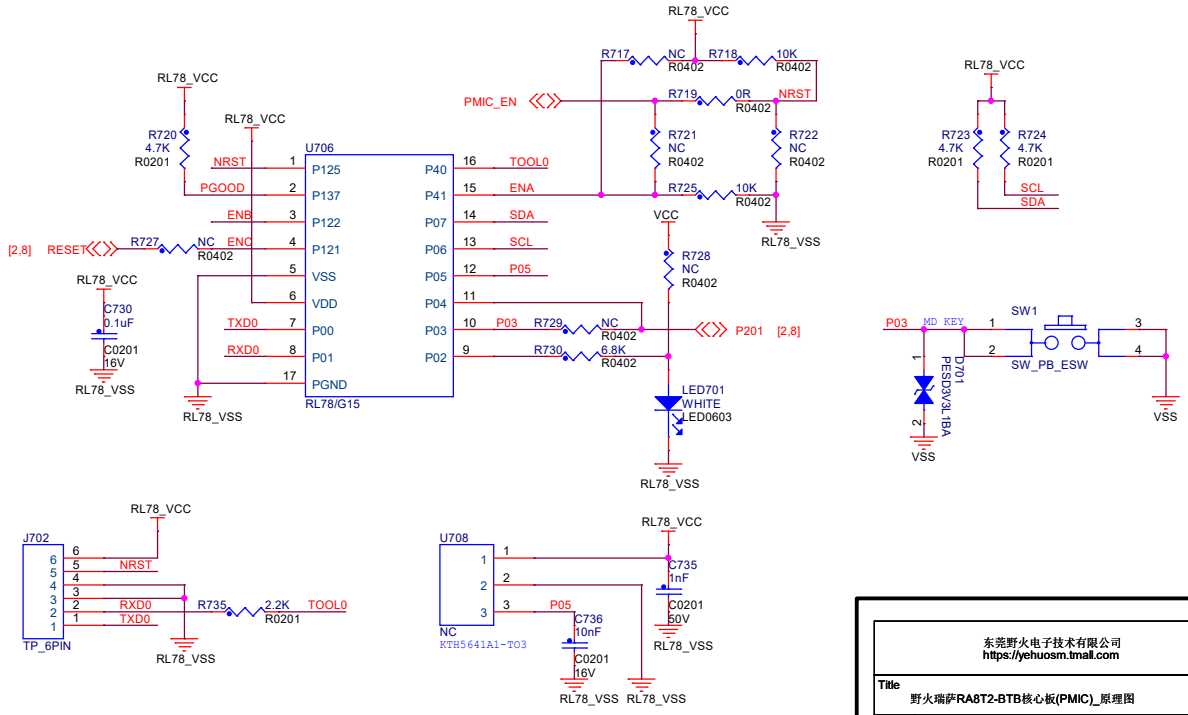
电源保护



PMIC



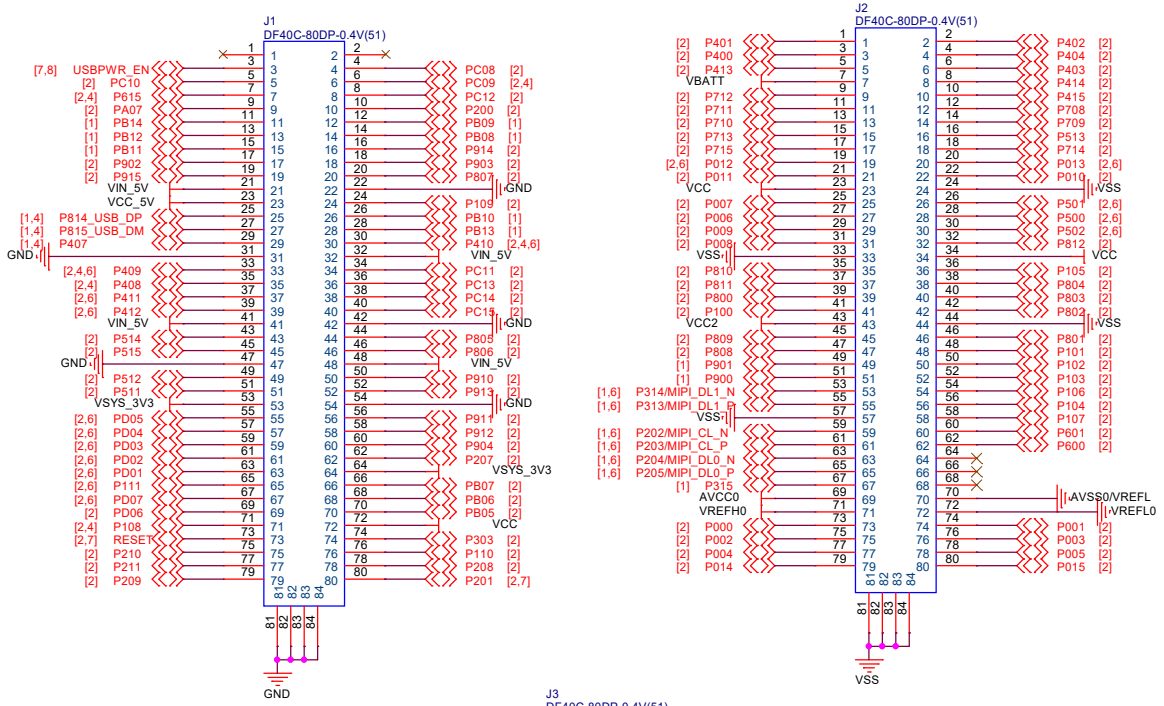
MCU

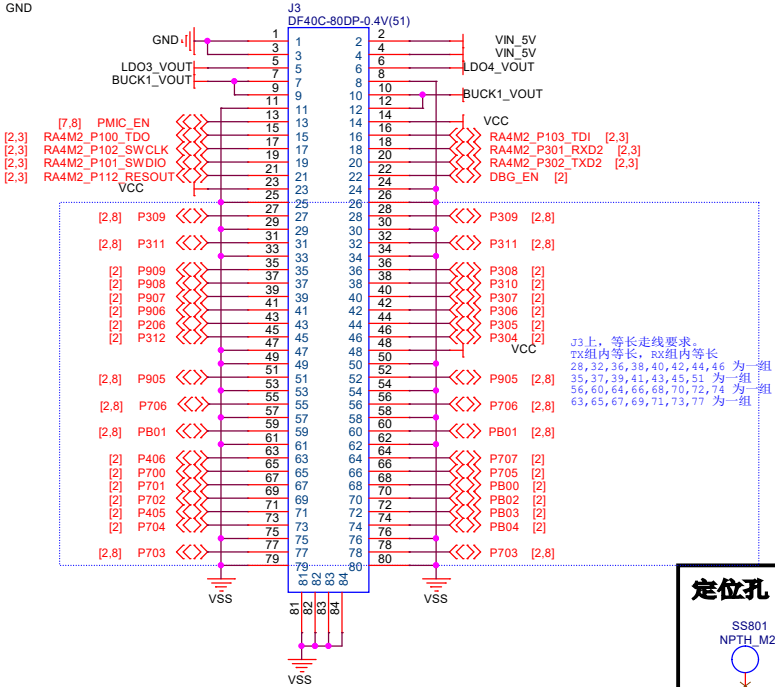


东莞野火电子技术有限公司 https://yehuosm.tmall.com		
Title: 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图		
Size: A3	Document Number: PMIC	Rev: V1.0
Date: Saturday, June 13, 2026	Sheet 7 of 10	

BTB座子

Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB





定位孔

SS801	SS802	SS803	SS804
NPTH_M2	NPTH_M2	NPTH_M2	NPTH_M2

东莞野火电子技术有限公司
https://yehuosm.tmall.com

Title	野火瑞萨RA8T2-BTB核心板(PMIC)_原理图	
Size A3	Document Number BTB	Rev V1.0
Date: Tuesday, June 09, 2026	Sheet 8 of 10	

历史版本

版本号	日期	设计	描述
V1.0	2026-06-08	LPC	初版发行

东莞野火电子技术有限公司 https://yehuosm.tmall.com		
Title 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图		
Size A4	Document Number 历史版本	Rev V1.0
Date: Tuesday, June 09, 2026	Sheet 9 of 10	

野火瑞萨RA8T2-BTB核心板(PMIC)_原理图_V1.0

目录

Page 1	MCU电源
Page 2	MCU I/O
Page 3	Jlink OB
Page 4	USB HS
Page 5	SDRAM/FLASH
Page 6	SD CARD/MIPI
Page 7	PMIC
Page 8	BTB
Page 9	历史版本
Page 10	目录
Page 11	
Page 12	
Page 13	
Page 14	

东莞野火电子技术有限公司
https://yehuosm.tmall.com

Title
野火瑞萨RA8T2-BTB核心板(PMIC)_原理图

Size	Document Number	Rev
A4	目录	V1.0

Date: Tuesday, June 09, 2026 Sheet 10 of 10